TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	51.8 ± 2.09
Correlation Stability	89.7 ± 0.46
Distribution Consistency	46.8 ± 5.02
Entropy Preservation	22.8 ± 1.60
Mean Z Score Drift	0.3 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	74.1 ± 5.11
Data Density	100.0 ± 0.00
Coverage Score	6.1 ± 2.47
Feature Accuracy	40.4 ± 0.80
Novelty Score	0.7 ± 0.02

TabuLLM: AI Benchmark Report

Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	3.000	-0.096	3.1%	OK
PRE4	3.282	3.242	-0.040	1.2%	OK
PRE5	4.569	2.518	-2.051	44.9%	Drift
PRE6	0.781	0.862	+0.081	10.4%	Warn
PRE7	0.066	0.000	-0.066	100.0%	Drift
PRE8	0.145	0.000	-0.145	100.0%	Drift
PRE9	0.066	0.000	-0.066	100.0%	Drift
PRE10	0.687	0.814	+0.127	18.4%	Warn
PRE11	0.166	0.008	-0.158	95.2%	Drift
PRE14	11.736	11.872	+0.136	1.2%	OK
PRE17	0.074	0.000	-0.074	100.0%	Drift
PRE19	0.004	0.000	-0.004	100.0%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	51.8 ± 2.09
Correlation Stability (*2)	89.7 ± 0.46
Distribution Consistency (*3)	46.8 ± 5.02
Entropy Preservation (*4)	22.8 ± 1.60
Mean Z Score Drift (*5)	0.3 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	74.1 ± 5.11

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	6.1 ± 2.47
Feature Accuracy (*10)	40.4 ± 0.80
Novelty Score (*11)	0.7 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8542553191489363	N/A	0.8521276595744683	-0.002127659574468077	
F1	N/A	N/A	N/A	0	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	N/A	0.847872340425532	-1.1102230246251566e-17	
F1	N/A	N/A	0.01111111111111111	0.01111111111111111	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	N/A	0.8553191489361703	0	
F1	N/A	N/A	N/A	0	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.21063829787234045	N/A	0.29680851063829783	0.08617021276595745	
F1	0.256855620661189	N/A	0.2659730768111307	0.009117456149941697	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	N/A	0.8191489361702127	0.005319148936170204	
F1	0.11972345964178996	N/A	0.12040298852521225	0.00067952888342228 95	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.808510638297872				

2

N/A	0.803191489361702	-0.00531914893617020

4

F1	0.11849989832593655	N/A	0.22102976925417317	0.10252987092823662	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8382978723404255	N/A	0.8404255319148938	0.00212765957446809 95	
F1	0.07594017094017094	N/A	0.19343473146407592	0.11749456052390499	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.852127659574468	N/A	0.8436170212765959	-0.008510638297872342	
F1	N/A	N/A	0.05232905982905982	0.05232905982905982	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	N/A	0.8404255319148938	-0.010638297872340451	
F1	0.037675637056442016	N/A	0.06419578357658853	0.026520146520146514	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	N/A	0.8553191489361703	0	
F1	N/A	N/A	N/A	0	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

TabuLLM: AI Benchmark Report

B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.733048087368364	2.3473150758542163	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.4335453080677496	3.2220098243040423	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.843563343691686	1.7242826014810595	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.6326642530487425	2.6305249195773643	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.511402158678218	1.7073111070580893	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	2.71749320530957	2.3236641500151274	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.5308991978335764	2.4517976144879983	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.2866047816101713	3.193290298703229	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	3.1749264743222367	3.5255724005310407	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.442385834366019	2.7609302827570614	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.004041312416173	2.5446531339156904	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	1.9325636741960373	2.113057501191905	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

TabuLLM: AI Benchmark Report

B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.894650886861196	2.1156200745944584	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.0343407161971974	2.0228241839923706	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	4.480376006113895	2.5496953831968274	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.803601000786675	2.111012890351562	1.0	0.0	0.0	0.0	1.0	1.0	12.0
5	3.0	2.166436772934634	2.150396815860071	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	3.026923614015270	2.638883884179848	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.946271347722624	1.9283876418846542	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	4.623469381880247	2.627309223121472	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	4.747324201695817	2.689105706410063	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.634967901776531	2.539920876682443	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.004025458645646	2.281526291526932	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.457496958067957	2.994308192367977	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

TabuLLM: AI Benchmark Report

B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.45011782690450 3	1.54433605203147 24	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	3.22390726666900 9	3.63728353910552 3	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.68020656665918 7	2.80769586110960 92	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	4.51842845307246 8	2.11053302197680 4	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.14136141546702 64	3.25864316431004 66	0.0	0.0	0.0	0.0	0.0	0.0	12.0
6	3.0	4.39703217811743 7	1.64472823854338 03	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.75773049228858 64	2.24944213106840 1	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	4.19893211284455 1	3.20768881665328 33	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.87419660420288 2	2.52839480494136 07	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.85790439576863 25	2.37953288373776 05	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	3.89789481658346 6	2.49501925225708 9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
6	3.0	3.26615168465515 6	1.49056902786647 07	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

TabuLLM: AI Benchmark Report

B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.77268185799048 1	2.99778594082722 54	1.0	0.0	0.0	0.0	1.0	1.0	12.0
2	3.0	2.33664478049940 35	2.63586516640394	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	3.02076847197024 23	2.90143533628331 6	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.53356494144696 94	2.94147530809050 6	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	3.0	3.28480357030834 7	2.54687803836204 77	0.0	0.0	0.0	0.0	0.0	0.0	12.0
6	3.0	2.57465032222036 25	2.71085215560264 3	1.0	0.0	0.0	0.0	1.0	1.0	12.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.18775090546053 97	2.17070406057946 07	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.66585087005810 76	2.32657291521828 75	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.59076466054015 55	2.94707876161635	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.10811510263127 6	2.24432981818047 23	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.95546001629491 96	2.56046312346570 07	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.30460938356233 85	2.04936148206151 2	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

TabuLLM: AI Benchmark Report

B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.73717032489737	2.96613056472549	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	4.42051583512153	2.92161973727576	0.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	3.06805940531955	1.36065743274279	0.0	0.0	0.0	0.0	0.0	0.0	11.0
4	3.0	3.78731173627688	3.43520593174644	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.58343203224502	2.09678813995871	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	4.07425468581667	3.33978800092305	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.93632051619314	2.46753087289482	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	1.89354984583581	1.99411609498805	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.54636439327630	2.14110290507812	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.72546316207188	3.31448044222390	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	4.56145828020041	2.41426580742566	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.80102175619730	1.95771610942550	1.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0